

Linear 2-DOF Bicycle Model

1 Concept

The 2-DOF linear bicycle model reduces the vehicle to one mass, two axles, and linear tyre forces. The motion variables of interest are forward velocity u , lateral velocity v , and yaw rate r . With $u = V$ held constant, the model becomes a linear state-space system in v and r . It is the reference model used in ESC and the standard first question in a vehicle-dynamics interview.

2 Assumptions

- No lateral load transfer (single-track, hence “bicycle”).
- No longitudinal load transfer.
- No rolling or pitching motions of the body.
- Linear-range tyres.
- Constant forward velocity V , chosen by the user.
- No aerodynamic effects.
- Position control.
- No chassis or suspension compliance.
- Smooth road, no ride motions.
- Small-angle assumption, in radians:

$$\sin \theta \approx \theta, \quad \cos \theta \approx 1.$$

3 Motion Variables and Lateral Acceleration

$$u = V = \text{constant forward velocity}, \quad v = \text{lateral velocity}, \quad r = \text{yaw rate}.$$

In the vehicle axis system the lateral acceleration is

$$a_y = Vr + \dot{v}.$$

With the sideslip angle $\beta = v/V$ and V constant,

$$a_y = Vr + \dot{v} = Vr + V\dot{\beta} = V(r + \dot{\beta}).$$

The first term is the centrifugal term; the second is the direct lateral acceleration term.

4 Slip Angles

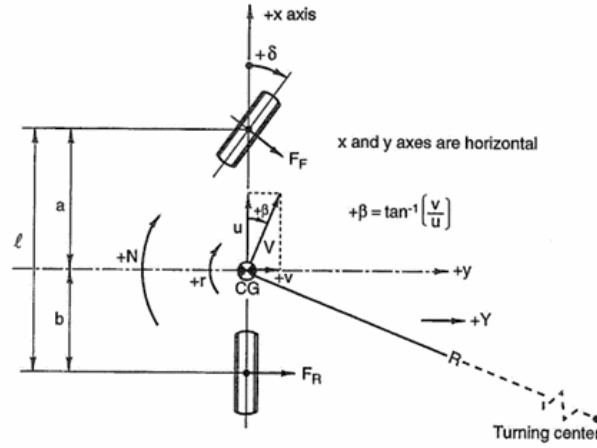


Figure 1: 2 DOF bicycle model.

4.1 Rear tyre

If the vehicle experiences a positive yaw rate r about the vertical z -axis, the rear tyre sees a lateral velocity component $-br$, where b is the distance from the CG to the rear axle. Hence

$$\alpha_R = \frac{v - br}{V} = \frac{v}{V} - \frac{br}{V} = \beta - \frac{br}{V}.$$

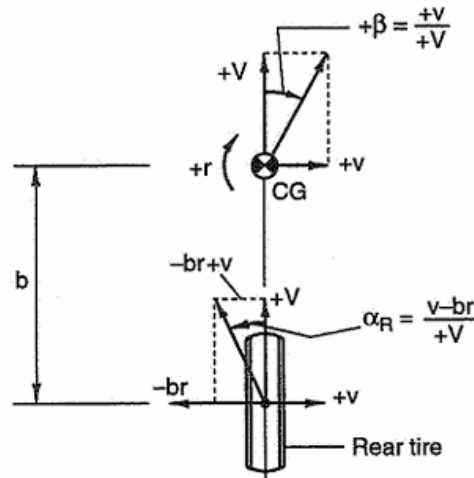


Figure 2: Rear tyre velocity and slip angle.

4.2 Front tyre

The front tyre slip angle has three components: vehicle sideslip β , yawing contribution ar/V , and steering angle δ :

$$\alpha_F = \frac{v + ar}{V} - \delta = \frac{v}{V} + \frac{ar}{V} - \delta = \beta + \frac{ar}{V} - \delta.$$

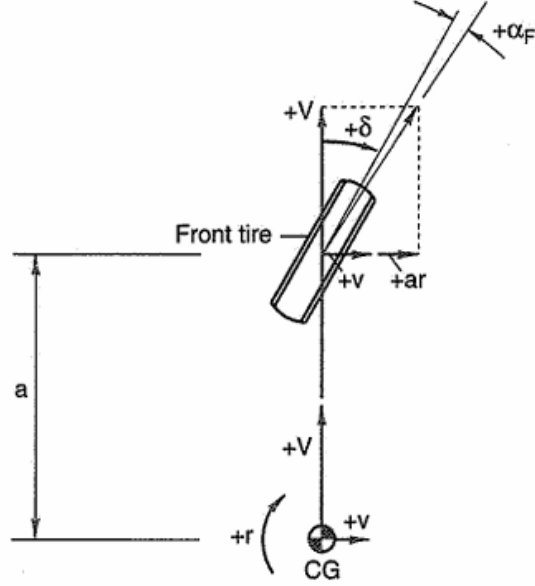


Figure 3: Front tyre slip angle.

5 Linear Tyre Forces

The lateral tyre forces vary linearly with slip angle:

$$Y_F = C_F \left(\beta + \frac{ar}{V} - \delta \right) = C_F \beta + C_F \frac{ar}{V} - C_F \delta,$$

$$Y_R = C_R \left(\beta - \frac{br}{V} \right) = C_R \beta - C_R \frac{br}{V}.$$

6 Total Lateral Force and Yaw Moment

$$Y = Y_F + Y_R = (C_F + C_R) \beta + \frac{1}{V} (aC_F - bC_R) r - C_F \delta.$$

$$N = N_F + N_R = aY_F - bY_R,$$

$$N = (aC_F - bC_R) \beta + \frac{1}{V} (a^2 C_F + b^2 C_R) r - aC_F \delta.$$

Y and N are the unbalanced forces and moments acting on the vehicle from the tyres. Since Y and N are linear in β , r , and δ , the principle of superposition holds and

$$Y = Y_\beta \beta + Y_r r + Y_\delta \delta, \quad N = N_\beta \beta + N_r r + N_\delta \delta.$$

7 Stability and Control Derivatives

Comparing the two forms gives the derivatives in terms of the front and rear tyre cornering stiffnesses C_F, C_R , the CG location a, b , and the forward velocity V :

$$\begin{aligned} Y_\beta &= C_F + C_R, & Y_r &= \frac{1}{V}(aC_F - bC_R), & Y_\delta &= -C_F, \\ N_\beta &= aC_F - bC_R, & N_r &= \frac{1}{V}(a^2C_F + b^2C_R), & N_\delta &= -aC_F. \end{aligned}$$

Derivative	Name	Nature
N_δ	Control Moment Derivative	Control
Y_δ	Control Force Derivative	Control
N_r	Yaw Damping Derivative	Damping
Y_β	Damping-in-Sideslip Derivative	Damping
N_β	Static Directional Stability	Coupling
Y_r	Lateral Force/Yaw Coupling	Coupling

Table 1: Derivatives for the simple 2-DOF automobile.

8 Equations of Motion

Combining the lateral acceleration with the force and moment expressions:

$$\begin{aligned} mV(r + \dot{\beta}) &= Y_\beta\beta + Y_rr + Y_\delta\delta, \\ I_z\dot{r} &= N_\beta\beta + N_rr + N_\delta\delta. \end{aligned}$$

9 State-Space Form

Take the state vector

$$X = \begin{bmatrix} v \\ r \end{bmatrix}, \quad \dot{X} = AX + B\delta.$$

Using $\beta = v/V$, $\dot{\beta} = \dot{v}/V$, and V constant, the lateral equation becomes

$$\dot{v} = \frac{C_F + C_R}{mV}v + \left(\frac{aC_F - bC_R}{mV} - V \right)r - \frac{C_F}{m}\delta,$$

and the yaw equation is

$$\dot{r} = \frac{aC_F - bC_R}{I_zV}v + \frac{a^2C_F + b^2C_R}{I_zV}r - \frac{aC_F}{I_z}\delta.$$

Therefore

$$A = \begin{bmatrix} \frac{C_F + C_R}{mV} & \frac{aC_F - bC_R}{mV} - V \\ \frac{aC_F - bC_R}{I_zV} & \frac{a^2C_F + b^2C_R}{I_zV} \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{C_F}{m} \\ -\frac{aC_F}{I_z} \end{bmatrix}.$$

10 Dimensional Analysis of A

Use SI units: C_F, C_R in N/rad, m in kg, I_z in kg m², a, b in m, V in m/s, v in m/s, r in rad/s, and radian is dimensionless.

Element	Expression	Units
A_{11}	$\frac{C_F + C_R}{aC_F - bC_R}$	$1/(\text{s} \cdot \text{rad}) = 1/\text{s}$
A_{12}	$\frac{mV}{aC_F - bC_R} - V$	$\text{m}/(\text{s} \cdot \text{rad}) = \text{m/s}$
A_{21}	$\frac{I_z V}{a^2 C_F + b^2 C_R}$	$1/(\text{m} \cdot \text{s} \cdot \text{rad})$
A_{22}	$\frac{I_z V}{a^2 C_F + b^2 C_R}$	$1/(\text{s} \cdot \text{rad}) = 1/\text{s}$
B_1	$-\frac{C_F}{m}$	$\text{m}/(\text{s}^2 \cdot \text{rad})$
B_2	$-\frac{aC_F}{I_z}$	$1/(\text{s}^2 \cdot \text{rad})$

Table 2: Units of the state-space matrix elements.

11 Testing with MATLAB

```
m = 1500;           % kg
Iz = 2500;          % kg*m^2
a = 1.2;             % m
b = 1.4;             % m
Caf = 80000;         % N/rad
Car = 90000;         % N/rad
Vx = 25;             % m/s
```

```
[A,B] = bicycle_model_matrices(p,Vx);
disp('A='); disp(A);
disp('B='); disp(B);
```

This prints approximately

$$A \approx \begin{bmatrix} 4.5333 & -25.8000 \\ -0.4800 & 4.6656 \end{bmatrix}, \quad B \approx \begin{bmatrix} -53.3333 \\ -38.4000 \end{bmatrix}.$$